LOW LEVEL DESIGN (LLD)

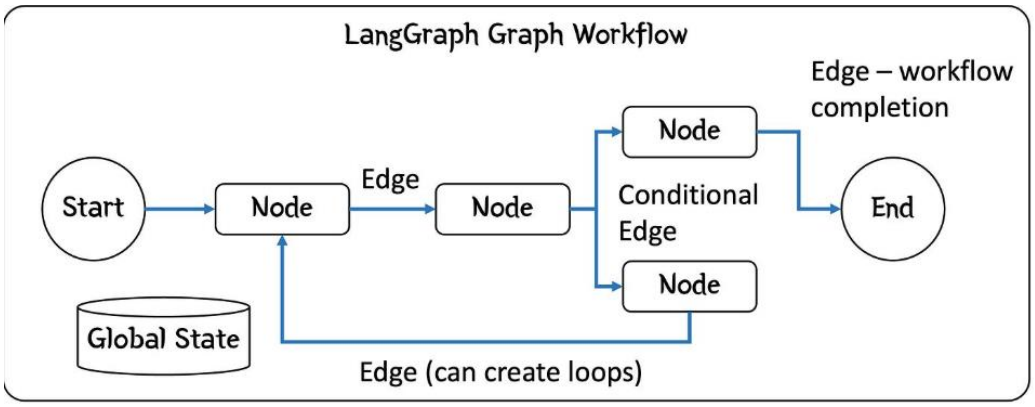
1. Modules

- Document Processing Module
- Chunking Module
- Embedding Module
- Vector Storage Module
- Retrieval Module
- Query Processing Module
- Graph Execution Module
- HITL Module

2. Data Structures

- Document → Raw text
- Chunk → Text segment
- Embedding → Vector
- Query → Input string
- State → Workflow data

3. Workflow Design



Flow:

- Input → Processing → Output

4. Conditional Routing

System checks:

- Confidence
- Context availability

If:

- Good → Answer
- Poor → Escalate

5. HITL Design

Triggered when:

- Low confidence
- Missing data

Process:

- Send to human
- Human responds
- Return answer

6. API Design

Input:

{

"query": "What is refund policy?"

}

Output:

{
"answer": "Refund processed within 7 days",
"confidence": "high"
}

7. Error Handling

- Missing PDF → error message
- No results → fallback response
- System failure → retry